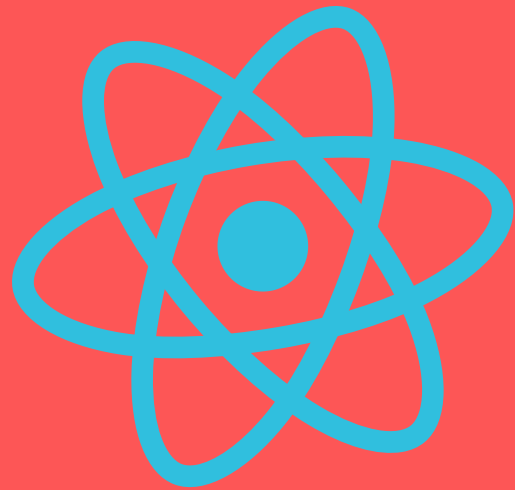
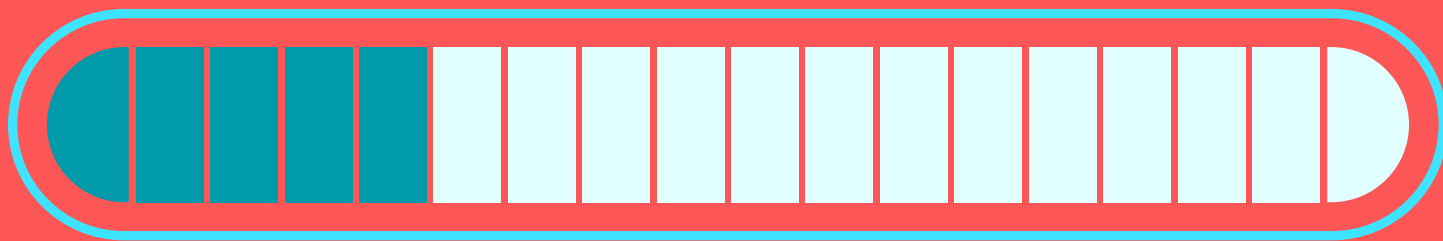


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What is Re-rendering?

How to Control It





What is Re-rendering?

Re-rendering is the process where React updates the UI based on state or prop changes.

Whenever a component's state updates, or it receives new props, it re-renders — along with its child components (unless optimized).





When Does Re-rendering Happen?

1. When useState or useReducer values change
2. When parent component re-renders (children re-render by default)
3. When props passed to a component change
4. When context values change and the component uses useContext





Why Re-rendering Can Be a Problem

In small apps, React handles re-rendering super efficiently.

But in large apps with:

- Expensive components (images, charts, etc.)
- Deep nested trees
- Repetitive updates

...unnecessary re-renders can hurt performance



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How to Optimize Re-rendering

Lets Discuss One By One





1. React.memo()

Wrap functional components to prevent re-rendering unless props actually change.

```
const MyComponent = React.memo(function  
MyComponent(props) {  
  return <div>{props.value}</div>;  
});
```





2. useCallback()

Prevents function props from being re-created on every render — useful when passing callbacks to child components.

```
const handleClick = useCallback(() => {  
  console.log("Clicked");  
}, []);
```





3. useMemo()

Memoizes computed values — useful for expensive calculations that shouldn't re-run unless dependencies change.

```
const filteredData = useMemo(() => {  
  return data.filter(item => item.active);  
}, [data]);
```





When to Use These Optimizations

Only optimize when:

- You've identified unnecessary renders
- Your component is slow/heavy
- You're passing functions or derived values deeply into children

Premature optimization can overcomplicate your code — always measure before optimizing!





Concept	Purpose
Re-rendering	Updates UI when state/props/context change
React.memo	Skips re-render if props didn't change
useCallback	Memoizes functions
useMemo	Memoizes calculated values